

GERA LEUMNITZ, Germany

WMO: 105670

Lat: 50.8814N

Lon: 12.1289E

Elev: 310

StdP: 97.66

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-14.5	-11.4	-16.6	0.9	-13.1	-13.7	1.2	-10.9	13.9	5.1	12.0	4.5	2.2	40	0.599

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	28.8	18.9	26.8	18.1	25.0	17.3	19.9	26.4	18.9	24.6	18.0	23.5	3.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.6	13.1	22.3	16.7	12.4	21.1	15.9	11.8	20.2	58.2	26.6	54.7	24.6	51.9	23.6	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.1	9.7	8.5	-16.6	32.6	6.8	1.5	-21.5	33.7	-25.5	34.6	-29.3	35.5	-34.2	36.6		
(5)			-16.9	22.0	6.6	1.3	-21.7	22.9	-25.5	23.7	-29.2	24.4	-34.0	25.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.5	-0.3	-0.8	3.7	7.2	12.1	15.1	17.6	17.6	13.9	9.3	3.9	1.4
	DBStd	7.70	5.55	5.65	4.39	3.81	3.65	3.51	3.21	3.36	3.29	3.99	3.88	4.09
	HDD10.0	1459	319	303	197	102	21	2	0	0	4	60	184	267
	HDD18.3	3683	578	536	452	335	193	107	54	54	137	279	433	526
	CDD10.0	897	1	1	3	16	87	156	235	236	121	39	2	1
	CDD18.3	79	0	0	0	0	1	11	31	32	4	1	0	0
	CDH23.3	791	0	0	0	1	22	120	305	294	47	2	0	0
	CDH26.7	184	0	0	0	0	0	24	76	80	4	0	0	0
(14) Wind	WSAvg	3.8	4.9	4.4	4.3	3.6	3.2	3.2	3.0	3.0	3.5	3.9	4.4	4.5
(15) Precipitation	PrecAvg	624	35	33	41	37	61	72	84	66	57	44	50	44
	PrecMax	854	69	56	97	80	149	129	140	180	131	91	129	91
	PrecMin	437	1	4	6	5	10	20	19	15	12	19	0	16
	PrecStd	109	17	15	24	19	31	27	33	40	27	19	29	21
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.9	13.9	17.6	21.5	25.4	29.5	31.1	32.2	27.2	22.6	14.7	11.9
		MCWB	8.4	9.6	10.6	12.8	16.5	19.2	20.4	19.8	17.4	15.7	10.8	9.1
	2%	DB	8.5	10.6	14.0	18.6	23.8	26.6	28.9	28.7	24.7	19.2	12.1	10.2
		MCWB	6.7	7.6	9.4	11.8	15.5	18.1	18.8	19.0	16.8	13.9	9.7	8.1
	5%	DB	7.3	8.0	11.3	16.0	21.5	24.3	26.8	26.5	22.3	16.9	10.5	8.7
		MCWB	5.7	5.8	7.9	10.5	14.5	17.2	18.3	17.9	16.0	12.9	8.6	7.1
	10%	DB	6.0	5.5	9.4	13.8	19.4	22.0	24.7	24.5	20.1	14.9	9.1	7.1
		MCWB	4.4	3.8	6.8	9.4	13.5	15.8	17.5	17.2	15.1	11.9	7.3	5.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.5	10.0	11.5	13.6	17.9	20.4	21.4	21.7	18.9	16.5	11.4	9.6
		MCDB	10.4	12.8	16.1	19.8	23.7	26.9	29.2	29.3	24.1	21.6	13.5	11.0
	2%	WB	7.0	7.7	9.7	12.2	16.2	18.7	20.1	19.7	17.6	14.7	10.0	8.3
		MCDB	8.4	10.0	13.5	17.1	21.8	24.7	26.4	26.6	22.9	18.2	11.9	10.0
	5%	WB	5.9	5.9	8.3	10.9	15.1	17.7	19.0	18.6	16.5	13.3	8.8	7.2
		MCDB	7.2	7.9	10.8	15.3	20.4	23.0	24.6	24.5	21.2	16.3	10.5	8.6
	10%	WB	4.5	4.0	7.1	9.6	14.0	16.6	17.9	17.8	15.5	12.0	7.5	5.8
		MCDB	5.8	5.5	9.3	13.2	18.5	21.1	23.4	23.1	19.3	14.3	9.0	7.1
(35) Mean Daily Temperature Range	5% DB	MDBR	5.0	6.2	7.0	8.5	9.8	9.2	10.2	10.1	8.9	7.9	5.8	4.3
		MCDBR	5.6	8.3	10.4	12.6	13.7	13.5	14.7	13.8	12.9	10.7	7.8	5.5
	5% WB	MCWBR	4.6	5.8	6.8	7.0	7.1	6.3	6.4	5.9	6.4	6.3	5.7	4.5
		MCDWR	5.4	8.2	9.6	11.5	12.1	12.1	12.8	11.9	11.3	9.8	7.2	5.3
(40) Clear-Sky Solar Irradiance	taub		0.305	0.328	0.370	0.405	0.408	0.406	0.422	0.413	0.379	0.355	0.328	0.300
		taud	2.390	2.351	2.270	2.218	2.242	2.273	2.234	2.276	2.355	2.405	2.418	2.392
	Ebn at Noon		722	798	822	833	848	851	829	816	806	756	685	667
		Edh at Noon	62	83	109	128	131	128	131	120	99	79	61	54
(44) All-Sky Solar Radiation	RadAvg		0.77	1.51	2.52	4.02	4.85	5.24	4.97	4.30	3.09	1.81	0.86	0.59
	RadStd		0.13	0.23	0.34	0.52	0.58	0.44	0.59	0.41	0.42	0.22	0.12	0.07

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (54 neighbors)		+0.39	N/A	N/A	N/A	+0.33	+0.37	N/A	N/A	N/A	+23	

Nomenclature: See separate page